torch.nn.utils.spectral_norm#

https://pytorch.org/docs/stable/generated/torch.nn.utils.spectral_norm.html#

Description:

Apply spectral normalization to a parameter in the given module. Spectral normalization stabilizes the training of discriminators (critics) in Generative Adversarial Networks (GANs) by rescaling the weight tensor with spectral norm σ\sigmaσ of the weight matrix calculated using power iteration method. If the dimension of the weight tensor is greater than 2, it is reshaped to 2D in power iteration method to get spectral norm. This is implemented via a hook that calculates spectral norm and rescales weight before every forward() call. See Spectral Normalization for Generative Adversarial Networks . module (nn.Module) – containing module name (str, optional) – name of weight parameter

Function Signature

```
torch.nn.utils.spectral_norm(module, name='weight',
n_power_iterations=1, eps=1e-12, dim=None)[source]#

Parameters

Returns

Return type
```

Returns:

T_module

Code Examples

```
>>> m = spectral_norm(nn.Linear(20, 40))
>>> m
Linear(in_features=20, out_features=40, bias=True)
>>> m.weight_u.size()
torch.Size([40])
```

Notes & Warnings

> **NOTE**
>
> Note This function has been reimplemented as torch.nn.utils.parametrizations.spectral_norm() using the new parametrization functionality in torch.nn.utils.parametrize.register_parametrization(). Please use the newer version. This function will be deprecated in a future version of PyTorch.

Detailed Documentation

Documentation

Apply spectral normalization to a parameter in the given module.

Spectral normalization stabilizes the training of discriminators (critics) in Generative Adversarial Networks (GANs) by rescaling the weight tensor with spectral norm σ of the weight matrix calculated using power iteration method. If the dimension of the weight tensor is greater than 2, it is reshaped to 2D in power iteration method to get spectral norm. This is implemented via a hook that calculates spectral norm and rescales weight before every forward() call.

See Spectral Normalization for Generative Adversarial Networks .

module (nn.Module) – containing module

name (str, optional) – name of weight parameter n_power_iterations (int, optional) – number of power iterations to calculate spectral norm

eps (float, optional) – epsilon for numerical stability in calculating norms

dim (int, optional) – dimension corresponding to number of outputs, the default is 0, except for modules that are instances of ConvTranspose{1,2,3}d, when it is 1

The original module with the spectral norm hook

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